

When can we use AIC and why should we not?

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What AIC Is For

- Key idea: AIC is a criterion for comparing statistical models fit to the same dataset

Interpretation

- $-2 \log(L)$: measures lack of fit (lower = better fit)
- $2k$: penalty for model complexity (more parameters = larger penalty)

Number of model parameters

$$AIC = 2k - 2\ln(\mathcal{L})$$

$\mathcal{L} = \mathcal{L}(\hat{\theta})$ = maximum value of the likelihood function of the model

The diagram shows the AIC formula with two callouts. A purple callout points to the 'k' in '2k', indicating it represents the number of model parameters. A red callout points to the 'ln(L)' part of the formula, indicating it represents the maximum value of the likelihood function of the model.

What AIC is trying to estimate

Goal:

- choose the model with the best out-of-sample predictive performance
- Approximates performance on new data, not just fit on the current sample (New data is generated from the same data-generating process)
- Comparisons assume the models are within the same statistical framework
- Lower AIC means better expected generalization

Assumptions Behind AIC

AIC comparisons are valid only if:

- Same response data : likelihood is evaluated on the same outcome
- Same observations : $\log L$ must be computed on the same set of data points
- Models fitted by (approximate) MLE : AIC is derived under maximum likelihood (or a close approximation)
- Number of parameters k is well-defined: Parameters are counted consistently across models

When Can We Use AIC?

- Same dataset (same n): Likelihood is computed on the same observations
- Same outcome variable
- Same likelihood family / same statistical framework
- Only model structure differs

Examples (good AIC comparisons):

- Add/remove covariates
- Different random effect structures
- SSM (random walk, 24h) vs SSM (correlated random walk, 24h)

From Valid
To Misuse
(Our Whale shark Project)

When AIC Fails: Our Whale Shark Project

Research Context

- Irregular satellite telemetry data
- SSM is used to regularize the observations before MPM

Project Workflow

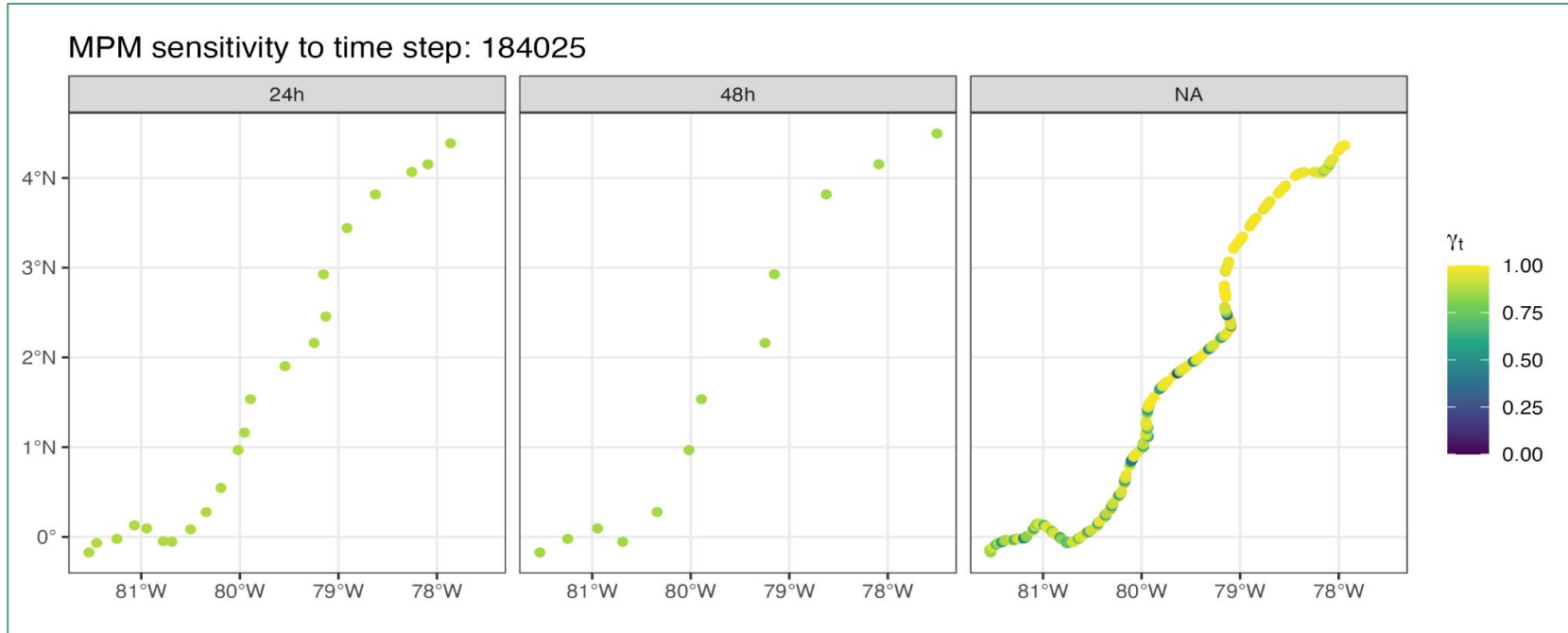
Irregular observations → SSM regularization → Regularized locations → MPM fitting
(choose time step:
24 vs 48 hours ?)

Key Decision

24h VS 48h is chosen before MPM model fitting

When AIC fails : Our Whale Shark Project

Same raw data
Different time steps



When AIC fails : Our Whale Shark Project

The problem

- SSM at different time steps creates different dataset
- MPM likelihood computed on different numbers of observations
- $AIC = -2\log L + 2k \leftarrow \log L$ not comparable across different sample sizes

What we should have done

- Choose time steps based on:
 - Whale shark movement ecology (how often do they change direction?)
 - GPS tag sampling frequency and accuracy
 - Research question scale
- Not a statistical model selection problem

From Valid
To Misuse
(General Examples)

When AIC fails: Penalized Models

Example: LASSO Regression

How Lasso works

- Minimize $-\log L(\beta) + \lambda \sum |\beta_j|$
- Penalty λ shrinks coefficients (some to exactly 0)
- Acts as automatic variable selection

Why AIC fails

- Not pure MLE on L
- Effective k not defined
- Model changes with tuning

Solution: Cross-validation

When AIC fails: Design Decisions

The pattern: data construction $\neq$ model selection

Examples

- Time resolution: 24h vs 48h (Our whale shark case)
- Spatial scale: fine-grained vs coarse grid
- Aggregation: raw vs summarized data

Why AIC is invalid

- The choice defines dataset
- Different dataset $\rightarrow$ different likelihoods $\rightarrow$ not comparable

Solution: Use domain knowledge; sampling limitations

When AIC fails: Transformed Response Variables

Example: Y vs $\log(Y)$

The mistake

- Model 1: $Y \sim X \rightarrow AIC_1 = 450$
- Model 2: $\log(Y) \sim X \rightarrow AIC_2 = 120$
- Lower AIC_2 means better model? 

Why this fails

- Y and $\log(Y)$ are different response variables
- Likelihood on different scales $\rightarrow$ AIC not comparable

Solution: Choose transformation first (based on theory), then compare models

AIC works when:

- Same dataset
- Same response variable
- True (unpenalized) likelihood

AIC fails when:

- Data are constructed differently
 - Estimation is penalized or quasi-likelihood
 - Response variables differ
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